1 Week 10

Definition 1.1. If $E \in \mathcal{E}, f : E \to \overline{\mathbb{R}}$ then we say that f is Lebesgue measurable if $f^{-1}(B)$ is Lebesgue measurable for all $B \in \mathcal{B}(\overline{\mathbb{R}})$.

Lemma 1.2. If $f : E \to \overline{\mathbb{R}}$ and if we define

$$\tilde{f}(x) = \begin{cases} f(x) & x \in E \\ 0 & x \notin E \end{cases}$$

then f is Lebesgue measurable if and only if $\tilde{f}$ is Lebesgue measurable.

Proof. If $B \in \mathcal{B}(\overline{\mathbb{R}})$ then if $0 \notin B$, $f^{-1}(B) = \tilde{f}^{-1}(B)$ and if $0 \in B$ then $\tilde{f}^{-1}(B) = f^{-1}(B) \cup E^c$, hence $f^{-1}(B) = f^{-1}(B) \cap E$. $\square$

Definition 1.3. We define $\mathbb{1}_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$. If it is Lebesgue measurable then so is E.

If $E \in \mathcal{E}$ we say that $\varphi : E \to \overline{\mathbb{R}}$ is simple if $\varphi = \sum_{k=1}^{n} \alpha_k \mathbb{1}_{E_k}$ where $E_k \subseteq E$ are Lebesgue measurable and $\alpha_k \in \mathbb{R}$.

Note that you can always rewrite the representative so that the E_k are disjoint and the α_k's are unique. This representatation is called canonical.

Lemma 1.4. Let $f : E \to \mathbb{R}$ be a bounded Lebesgue measurable function with $E \in \mathcal{E}$, and let $\varepsilon > 0$. Then there exist simple functions φ_ε and ψ_ε such that $\varphi_\varepsilon \leq f \leq \psi_\varepsilon$ and $|\psi_\varepsilon(x) - \varphi_\varepsilon(x)| < \varepsilon$ for all $x \in E$.

Proof. Suppose that $M > |f(x)|$ for all $x \in E$. Let $y_i \in \mathbb{R}$ be an increasing sequence satisfying $y_1 = -M$ and $y_n = M$. Then we let $E_k = f^{-1}((y_k, y_{k+1}))$ for all $1 \leq k \leq n-1$. We set $\varphi = \sum_{k=1}^{n-1} y_k \mathbb{1}_{E_k}$ and $\psi = \sum_{k=1}^{n-1} y_{k+1} \mathbb{1}_{E_k}$. By definition $\varphi \leq f \leq \psi$ and $|\varphi - \psi| \leq \max_k |y_{k+1} - y_k|$. By choosing y_k to be close enough, we get the second desired condition. $\square$

Theorem 1.5. Let $f : E \to \overline{\mathbb{R}}$ be Lebesgue measurable. Then

(i) If $f \geq 0$ then there exists an increasing sequence of simple functions $\varphi_n \geq 0$ such that $\lim_{n\to\infty} \varphi(x) = f(x)$ for all $x \in E$.

(ii) In general, there exists a sequence of simple functions φ_n such that $\lim_{n\to\infty} \varphi_n(x) = f(x)$ and $|\varphi_n(x)| \leq |f(x)|$ for all $x \in E$.

Proof. (ii) follows from (i) by writing $f = f_+ - f_-$ and finding simple ψ_n, χ_n that approximate f_+, f_- in the way of (i) and defining $\varphi_n = \psi_n - \chi_n$. Since at most one of f_+, f_- is non-zero at a given x, the same is the case for ψ_n, χ_n and hence the absolute value condition holds. The pointwise convergence obviously holds as well.

For (i), if $f \geq 0$, if we set $f_n = \max\{f, n\}$ then by the previous lemma, there is an increasing sequence $\psi_n \leq f_n$ and $|f_n - \psi_n| < 1/n$ for all n. Then we take $\varphi_1 = \max\{\psi_1, 0\}$ and $\varphi_k = \max\{0, \psi_k, \varphi_{k-1}\}$ for $k \geq 2$. By induction, we find that φ_k's are step functions such that $0 \leq \varphi_k \leq \varphi_{k+1} \leq f_k$, and $|f_k - \varphi_k| < 1/k$. With this we get $\varphi_n(x) \to f(x)$ for all $x \in E$. $\square$

Theorem 1.6 (Egorov)**.** Let $f_n : E \to \mathbb{R}$ be a sequence of Lebesgue measurable functions converging pointwise to $f : E \to \mathbb{R}$. We also assume that $m(E) < \infty$. Then for all $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $m(E \setminus F) < \varepsilon$ and $f_n \to f$ converges uniformly on F (that is, for all $\delta > 0$ there is some N such that $|f(x) - f_n(x)| < \delta$ for all $x \in F$ and $n \geq N$).

Lemma 1.7. In the setting of the theorem, for all $u > 0$ and $\delta > 0$ there is some $A \subseteq E$ and an $N \geq 0$ such that $m(E \setminus A) < \delta$ and for all $x \in A$ and $n \geq N$ we have $|f_n(x) - f(x)| < u$.

Proof. Set $E_n = \{x \in E : |f_n(x) - f(x)| < u, \forall k \geq n\} = \cap_{k=n}^{\infty}\{x : |f_k(x) - f(x)| < u\}$. Since $f_n \to f$ pointwise, $E = \cup_{n=1}^{\infty} E_n$. Furthermore, E_n are increasing ($E_n \subseteq E_{n+1}$ for all n). We then have $m(E) = \lim_{n\to\infty} m(E_n)$. Since $m(E) < \infty$, there is some N such that $|m(E) - m(E_N)| < \delta$, and hence $m(E \setminus E_N) = m(E) - m(E_N) < \delta$. Taking $A = E_N$ completes the proof. □

Proof (Egorov). By the lemma, for all n there is some A_n such that $m(E \setminus A_n) < \varepsilon/2^n$ and there exists $M(n)$ such that $|f(x) - f_k(x)| < 1/n$ for all $k \geq M(n)$. Setting $A = \cup_{n=1}^{\infty} A_n$, we have

$$\begin{aligned} m(E \setminus A) &= m(E \cap A^c) \\ &= m\left(E \cap \left(\bigcup_{n=1}^{\infty} A_n^c\right)\right) \\ &= m\left(\bigcup_{n=1}^{\infty} E \setminus A_n\right) \\ &\leq \sum_{n=1}^{\infty} m(E \setminus A_n) \\ &\leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} \\ &= \varepsilon \end{aligned}$$

Now if $x \in A$ then for all n, $x \in A_n$, so for all $k \geq M(n)$, we have $|f_k(x) - f(x)| < 1/n$. We thus see that $f_k \to f$ uniformly on A.

We now take F closed such that $m(A \setminus F) \leq \varepsilon$ with $F \subseteq A$ to get $m(E \setminus F) \leq m(E \setminus A) + m(A \setminus F) \leq 2\varepsilon$. □

Theorem 1.8 (Lusin)**.** Let $f : E \to \mathbb{R}$ be Lebesgue measurable with $m(E) < \infty$. Then for all $\varepsilon > 0$ there is a closed set F and a continuous $g : F \to \mathbb{R}$ such that $m(E \setminus F) < \varepsilon$ and $f = g$ on F.

Proposition 1.9. Let $\varphi : E \to \mathbb{R}$ be a simple function and let $\varepsilon > 0$. Then there exists a closed $F \subseteq E$ and a continuous $g : \mathbb{R} \to \mathbb{R}$ such that $m(E \setminus F) < \varepsilon$ and $g = \varphi$ on F.

Proof. Write $\varphi = \sum_{k=1}^{\infty} \alpha_k \mathbb{1}_{E_k}$ with E_k being disjoint. Take F_k closed and bounded such that $F_k \subseteq E_k$ with $m(E_k \setminus F_k) < \varepsilon/n$. Since F_k's are disjoint compact sets, we have $\delta = \min_{i\neq j} \inf_{x\in F_i, y\in F_j} |x - y| > 0$. For any set A, define $d(x, A) = \inf_{y\in A} |x - y|$.

Exercise 1.10. Show that $x \mapsto d(x, A)$ is continuous.

Proof. TODO □

Let $\rho(x) : \mathbb{R}_{\geq 0} \to \mathbb{R}$ as 1 on $[0, \delta/3]$, 0 on $[2\delta/3, \delta]$, and linear on $[\delta/3, 2\delta/3]$ such that ρ becomes connected. Define $g_i = \alpha_i \rho(d(x, F))$. Each g_i is continuous and if $x \in F_i$ then $\rho(d(x, F_i)) = 1$ and if $x \in F_j$ for $j \neq i$ then $\rho(d(x, F_j)) = 0$ because $d(x, F_i) \geq \delta$. We then define $g(x) = \sum_{i=1}^{n} g_i(x)$.

Define $F = \bigcup_{n=1}^{\infty} F_n$, and note that $g(x) = \varphi(x)$ for $x \in F$ and $m(E \setminus F) \leq \sum_{k=1}^{n} m(E_k \setminus F_k) \leq n\varepsilon/n = \varepsilon$.

Lemma 1.11. If E has finite measure then for any $\varepsilon > 0$ there is some compact K such that $m(E \setminus K) < \varepsilon$.

Proof. Set $E_n = E \cap [-n, n]$. Then since $\bigcup_n E_n = E$ and $m(E) = \lim_{n\to\infty} m(E_n)$, we can find E_k such that $m(E \setminus E_k) < \varepsilon$. □

Then take $F \subseteq E_k$ with F closed and $m(E_k \setminus F) < \varepsilon \Rightarrow m(E \setminus F) \leq m(E \setminus E_k) - m(E_k \setminus F_k) \leq 2\varepsilon$. □

Proof (Lusin). Take φ_n simple functions such that φ converge to f pointwise and let g_n be a sequence of continuous functions such that $g_n = \varphi_n$ on a closed set F_k with $m(E \setminus F_n) \leq \varepsilon/2^n$. By Egorov, there is a closed $G \subseteq E$ with $m(E \setminus G) < \varepsilon$ such that $\varphi_n \to f$ uniformly on G. Take $F = (\bigcap_{n=1}^{\infty} F_n) \cap G$. Then F is closed and $m(E \setminus F) \subseteq m(E \setminus G) + \sum_n m(E \setminus F_n) \leq 2\varepsilon$. On F, we have $g_n = \varphi_n$ for all n, so $g_n \to f$ uniformly on F. Thus the funciton $g(x) = \lim_{n\to\infty} g_n(x)$ for $x \in F$ in continuous. □

2 Week 11

Definition 2.1. We define

$$\int_E \sum_{j=1}^{n} \alpha_j \mathbb{1}_{E_j} = \sum_{j=1}^{n} \alpha_j m(E_j)$$

and

$$\int_E f = \sup \left\{ \int_E \varphi : \varphi \leq f, \varphi \text{ simple} \right\}$$

Theorem 2.2 (Monotone Convergence Theorem)**.** Let $f_n : E \to [0, \infty]$ be measurable, $f_n \leq f_{n+1}$ for all n. Let $f = \lim_{n\to\infty} f_n$ pointwise. Then $\lim_{n\to\infty} \int_E f_n = \int_E f$.

Proof. Since $f_n \leq f$, we have $\int_E f_n \leq \int_E f \Rightarrow \lim_{n\to\infty} \int_E f_n \leq \int_E f$. Let $0 < \alpha < 1$ and let $\varphi : E \to [0, \infty)$ be simple with $\varphi \leq f$. Suffices to show that $\lim_{n\to\infty} \int_E f_n \geq \alpha \int_E \varphi$. Let $E_k = \{x \in E : f_k(x) \geq \alpha\varphi(x)\}$ and note that $E_k \subseteq E_{k+1}$ and $E = \bigcup_{k=1}^{\infty} E_k$. We then have

$$\lim_{n\to\infty} \int_E f_n \geq \int_E f_k \geq \int_E \mathbb{1}_{E_k} f_k \geq \int_E \mathbb{1}_{E_k} \alpha\varphi = \alpha \int_E \mathbb{1}_{E_k} \varphi$$

Writing $\varphi = \sum_{j=1}^{n} \alpha_j \mathbb{1}_{F_j}$, we have

$$\alpha \int_E \mathbb{1}_{E_k} \varphi = \alpha \sum_{j=1}^{n} \alpha_j m(F_j \cap E_k) \leq \lim_{n\to\infty} \int_E f_n$$

This holds for all k so take $k \to \infty$. Since $F_j \cap E_k \subseteq F_j \cap E_{k+1}$ for all k and $\bigcup_{k=1}^{\infty} F_j \cap E_k = F_j$ so by continuity of measure $\lim_{n\to\infty} m(F_j \cap E_n) = m(F_j)$. Thus we have $\lim_{n\to\infty} \int_E f_n \geq \alpha \sum_{j=1}^{n} \alpha_j m(F_j) = \alpha \int_E \varphi$. Taking $\alpha \to 1$ and using the definiion of Lebesgue integral, we see that $\lim_{n\to\infty} \int_E f_n \geq \int_E f$. Hence the equality holds. □

Lemma 2.3. If $f, g : E \to [0, \infty]$ are measurable, then $\int_E f + g = \int_E f + \int_E g$.

Proof. Since f, g are nonnegative and measurable, there exist nonnegative simple functions φ_n, ψ_n that are increasing to f, g and $\varphi_n \to f, \psi_n \to g$ pointwise. Then

$$\int_E f + g = \int_E \lim_{n\to\infty} (\phi_n + \psi_n) = \lim_{n\to\infty} \int_E \phi_n + \psi_n = \lim_{n\to\infty} \int_E \phi_n + \lim_{n\to\infty} \int_E \psi_n = \int_E f + \int_E g$$

□

Corollary 2.4. If $f_n : E \to [0, \infty]$ are measurable then $\int_E \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_E f_n$

Proof. Just apply MCT. □

Proposition 2.5. Let $f : E \to [0, \infty]$ be measurable. Then $\int_E f = 0$ iff $f = 0$ a.e. on E.

Proof. If $f = 0$ a.e. on E then if $\varphi = \sum_{j=1}^{n} \alpha_j \mathbb{1}_{E_j}$ with E_j disjoint, $\alpha_j \geq 0$, satisfies $\varphi \leq f$, then if $\alpha_j > 0$, we have $m(E_j) = 0$. Thus $0 \leq \int_E f \leq \int_E \varphi \leq 0$ so $\int_E f = 0$.

If $\int_E f = 0$, consider the set $f^{-1}((0,\infty]) = \bigcup_{k=1}^{\infty} f^{-1}((1/k,\infty])$, and define $E_k := f^{-1}((1/k,\infty])$. We then have

$$0 = \int_E f \geq \int_E \mathbb{1}_{E_k} f \geq \int_E \frac{1}{k} \mathbb{1}_{E_k} = \frac{m(E_k)}{k} \geq 0$$

Thus $m(E_k) = 0$ so $m(f^{-1}((0,\infty])) = \lim_{n\to\infty} m(E_k) = 0$. Since the set on which $f > 0$ is of measure zero, we have $f = 0$ a.e. on E. □

Theorem 2.6 (Better MCT)**.** Let $f_n, f : E \to [0,\infty]$ be measurable. Assume that $f_n \leq f_{n+1}$ a.e. and assume $f = \lim_{n\to\infty} f_n$ a.e. on E. Then $\lim_{n\to\infty} \int_E f_n = \int_E f$

Proof. Let $E_0 = \{x \in E : \lim_{n\to\infty} f_n(x) = f(x), f_n(x) \leq f_{n+1}(x)\}$. Then $m(E \setminus E_0) = 0$ ($E \setminus E_0$ is a union of sets of measure zero). Now, we have

$$\lim_{n\to\infty} \int_E f_n = \lim_{n\to\infty} \int_E f_n(\mathbb{1}_{E_0} + \mathbb{1}_{E\setminus E_0}) = \lim_{n\to\infty} \int_E f_n \mathbb{1}_{E_0} + \int_E f_n \mathbb{1}_{E\setminus E_0} = \int_E \lim_{n\to\infty} f_n \mathbb{1}_{E_0} = \int_{E_0} f = \int_E f$$

□

Theorem 2.7 (Fatou's lemma)**.** If $f_n : E \to [0,\infty]$ are measurable then

$$\int_E \liminf_{n\to\infty} f_n \leq \liminf_{n\to\infty} \int_E f_n.$$

In particular, if $f_n \to f$ a.e. then $\int_E f \leq \liminf_{n\to\infty} \int_E f_n$.

Proof.

$$\int_E \liminf_{n\to\infty} f_n = \int_E \lim_{n\to\infty} \left(\inf_{k\geq n} f_k \right) = \lim_{n\to\infty} \int_E \inf_{k\geq n} f_k \leq \lim_{n\to\infty} \inf_{k\geq n} \int_E f_k = \liminf_{n\to\infty} \int_E f_n$$

□

Theorem 2.8 (Markov's inequality)**.** If $f : E \to [0,\infty]$ is measurable then for all $\alpha > 0$, $m(f^{-1}([\alpha,\infty])) \leq \frac{1}{\alpha} \int_E f$.

Proof. Let $F = f^{-1}([\alpha,\infty])$, then $f(x)/\alpha \geq 1$ on F, so

$$m(F) = \int_E \mathbb{1}_F \leq \int_E \mathbb{1}_F \frac{f}{\alpha} \leq \frac{1}{\alpha} \int_E f$$

□

Corollary 2.9. If $t > 0$, $m(f^{-1}([\alpha,\infty])) \leq e^{-t\alpha} \int_E e^{tf}$.

Proof. Holds since $m(f^{-1}([\alpha,\infty])) = m((e^{tf})^{-1}([\alpha,\infty]))$. □

Corollary 2.10. If $\int_E f < \infty$ then $f(x)$ is finite a.e.

Proof. $m(f^{-1}(\infty)) \leq m(f^{-1}([n,\infty])) \leq \frac{1}{n} \int_E f \to 0$. □

2.1 Integral for general $f : E \to \overline{R}$

Definition 2.11. We say that a measurable $f : E \to \overline{R}$ is integrable if $\int_E f_+ + \int_E f_- = \int_E |f| < \infty$. If f is integrable then we define $\int_E f = \int_E f_+ - \int_E f_-$

Lemma 2.12. Integrable functions form a vector space in that if f, g are integrable functions then so is $f + g$ and if f is integrable then so is λf for any $\lambda \in \mathbb{R}$.

Proof. Note that $\int_E |f + g| \leq \int_E |f| + \int_E |g| < \infty$ and $\int_E |\lambda f| = |\lambda| \int_E |f| < \infty$ which proves the lemma. □

Exercise 2.13. Check that $\int_E \alpha f + \beta g = \alpha \int_E f + \beta \int_E g$ for all integrable f, g and constants α, β.

Exercise 2.14. Check that Fatou's lemma holds if $f_n \geq 0$ a.e.

Theorem 2.15 (Dominated Convergence Theorem)**.** Let $f_n : E \to \overline{R}$ be a sequence of measurable functions that converge pointwise a.e. to some $f : E \to \overline{R}$. Assume that there is an integrable g such that $|f_n| \leq g$ a.e. Then $\lim_{n\to\infty} \int_E f_n = \int_E f$.

Proof. By assumptions, $g + f_n, g - f_n$ are nonnegative a.e. so we can apply Fatou's lemma. This gives us

$$\int_E g + \int_E f = \int_E g + f = \int_E \lim_{n\to\infty} (g + f_n) \leq \liminf_{n\to\infty} \int_E g + f_n = \int_E g + \liminf_{n\to\infty} \int_E f_n$$

Similarly, we have

$$\int_E g - \int_E f \leq \int_E g + \liminf_{n\to\infty} \int_E -f_n = \int_E g - \limsup_{n\to\infty} \int_E f_n$$

Since $\int_E g = \int_E |g| < \infty$, we get

$$\int_E f \leq \liminf_{n\to\infty} \int_E f_n \leq \limsup_{n\to\infty} \int_E f_n \leq \int_E f$$

Thus we have $\lim_{n\to\infty} \int_E f_n = \int_E f$. □

Example 2.16. If E satisfies $m(E) < \infty$ and there is $B > 0$ s.t. $|f_n| \leq B$ a.e. then $\int_E \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int_E f_n$ by DCT with $g = B$.

Definition 2.17. If $f, g : E \to \overline{R}$ are measurable we say that $f \sim g$ if $f = g$ a.e. This is an equivalence relation. Let $\mathcal{F}$ be the set of equivalence classes and define $\alpha[f] + \beta[g] = [\alpha f + \beta g]$. It is easy to check that this gives a well-defined definition of addition and scalar multiplication. For $1 \leq p < \infty$, we define $L^p(E)$ to be the set of all $[f] \in \mathcal{F}$ such that $\|f\|_{L^p(E)} := \left(\int_E |f|^p\right)^{1/p} < \infty$.

3 Week 12

We will show that $C_c^\infty(\mathbb{R}) = L^p(\mathbb{R})$ for $1 \leq p < \infty$ and we will study Fourier series in L^2.

Definition 3.1. We say that $f : \mathbb{R} \to \overline{\mathbb{R}}$ is of compact support if there is some $M > 0$ such that $f(x) = 0$ a.e. for $x \in [-M, M]$.

Theorem 3.2. Smooth functions of compact support $C_c^\infty(\mathbb{R})$ are dense in L^p norm for $1 \leq p < \infty$.

We will only do the case for $p = 1$ for notational simplicity.

Lemma 3.3. Bounded functions of compact support are dense in L^1.

Proof. If $f \in L^1$, $f_n(x) = f(x)\mathbb{1}_{\{x:|f(x)|\leq n\}}\mathbb{1}_{|x|\leq n}$. We then have $f_n(x) \to f(x)$ a.e. since f is finite a.e. and for large enough n, the indicator functions become 1 where f is finite. Furthermore, we have $|f_n(x)| \leq |f(x)|$. We thus have that $|f_n(x) - f(x)| \to 0$ a.e. and $|f_n - f| \leq 2|f|$ where $\int_{\mathbb{R}} |f| < \infty$, hence by DCT, $\int_{\mathbb{R}} |f_n - f| \to 0$, meaning $\|f_n - f\|_1 \to 0$. □

Lemma 3.4. Simple functions with compact support are dense in L^1.

Proof. In order to approximate $f \in L^1$ by simple functions, we can assume that f is bounded with compact support. For all $\varepsilon > 0$, since f is bounded, there is a simple function φ such that $|\varphi(x) - f(x)| < \varepsilon$ a.e.

If $f(x) = 0$ a.e. $x \notin [-M, M]$ then $\psi(x) = \varphi(x)\mathbb{1}_{[-M,M]}$ then we also have $|\psi(x) - f(x)| < \varepsilon$ a.e. Then we have $\int_{\mathbb{R}} |\psi - f| = \int_{\mathbb{R}} |\psi - f|\mathbb{1}_{[-M,M]} \leq \varepsilon M$. Since M is a constant that depends only on f, we can take $\varepsilon \to 0$ to get the result. $\square$

Lemma 3.5. Let $E \subseteq [-M, M]$ for some $M > 0$ be measurable. Then there is a sequence of continuous functions of compact support f_n that approximate the indicator function $\mathbb{1}_E$ in L^1; that is $\int |\mathbb{1}_E - f_n| \to 0$.

Proof. We can find a closed $F \subseteq E$ such that $m(E \setminus F) < \varepsilon$, so we have $\int_{\mathbb{R}} |\mathbb{1}_E - \mathbb{1}_F| < \varepsilon$. It thus suffices to only consider the case that E is closed.

Let $\rho_n = x \mapsto \begin{cases} 1 & x \in [0, 1/n] \\ -n(x - 1/n) + 1 & x \in (1/n, 2/n) \\ 0 & x \in [2/n, \infty) \end{cases}$ and define $f_n(x) = \rho_n(d(x, F))$, where $d(x, F) = \inf_{y \in F} |x - y|$ is continuous. We thus have that f_n are continuous. Since F is closed, $d(x, F)$ is 0 if and only if $x \in F$, so $f_n(x) \to \mathbb{1}_F(x)$ for all $x \in \mathbb{R}$. Lastly, we have $|f_n(x)| \leq \mathbb{1}_{[-M-2,M+2]}$. This means that we have $|f_n - \mathbb{1}_F| \to 0$ pointwise and $|f_n - \mathbb{1}_F| \leq 2 \cdot \mathbb{1}_{[-M-2,M+2]}$, hence DCT applies to give us $\int_{\mathbb{R}} |f_n - \mathbb{1}_F| \to 0$. $\square$

Corollary 3.6. $C_c(\mathbb{R})$ is dense in L^1.

Lemma 3.7. For all $M > 0$ there is a smooth function ρ such that $\rho(x) = 1$ for $|x| \leq M$, $\rho(x) = 0$ for $|x| \geq M + 1$, and $0 \leq \rho \leq 1$

Proof. Define $\varphi(x) = x \mapsto \begin{cases} e^{-1/x} & x > 0 \\ 0 & x \leq 0 \end{cases}$.

Exercise 3.8. φ is smooth.

Proof. TODO $\square$

We then define smooth $\varphi_1(x)$ by

$$\varphi_1(x) = \frac{\varphi(x)\varphi(1 - x)}{\int_{\mathbb{R}} \varphi(x)\varphi(1 - x)}$$

and we define smooth φ_2 by

$$\varphi_2(x) = \int_{-\infty}^{x} \varphi_1(x)$$

Then $\rho(x) = \varphi_2(x + M \pm 1, 2)\varphi_2(M \pm 1, 2 - x)$ should work. $\square$

Lemma 3.9. If f is continuous and of compact support then there is a sequence of smooth functions of compact support f_n that approximate $f \in L^1$.

Proof. Since f is continuous and has support contained in $[-M, M]$, we can take a sequence of polynomials f_n such that $|f_n(x) - f(x)| \leq 1/n$ for all $x \in [-M - 2, M + 2]$. Then if ρ is as in the previous lemma, then $g_n(x) = \rho(x)f_n(x)$ then $|g_n(x) - f(x)| \leq 1/n$ for $x \in \mathbb{R}$ and $g_n(x) = 0$ for all $x \notin (-M - 1, M + 1)$. Thus we have $\int_{\mathbb{R}} |g_n - f| \to 0$. $\square$

We now consider $E = (-\pi, \pi)$, and we pretend that we have developed the theory of Lebesgue integration for $\mathbb{C}$-valued functions.

For $f \in L^2(E, \mathbb{C})$, we define $\mathcal{F}f : L^2(E) \to \ell^2(\mathbb{Z})$ by $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-inx} f(x)dx$. The function is well-defined by applying Cauchy-Schwarz:

$$\int_{\mathbb{R}} |f(x)|dx \le \left(\int_{-\pi}^{\pi} 1dx\right)^{1/2} \left(\int_{-\pi}^{\pi} |f(x)|^2 dx\right)^{1/2}$$

Lemma 3.10. $\mathcal{F}f \in \ell^2$ and $\mathcal{F}$ is a bounded linear operator: $\|\mathcal{F}f\| \le C\|f\|$.

Proof. There is a sequence of continuous bounded functions f_n that converge to f in L^2 and are compactly supported in $(-\pi, \pi)$ (by compactly supported here, we mean that the support is contained in some compact subset of $(-\pi, \pi)$). We then have

$$|\hat{f}(k) - \hat{f}_n(k)| \le \int_{-\pi}^{\pi} |f_n(x) - f(x)|dx$$

and by applying Cauchy-Schwarz, we get $|\hat{f}(k) - \hat{f}_n(k)| \le C\|f_n - f\|_{L^2}$. For all N, we have

$$\begin{aligned}
\sum_{|k|\le N} |\hat{f}(k)|^2 &= \lim_{n\to\infty} \sum_{|k|\le N} |\hat{f}_n(k)|^2 \\
&\le \lim_{n\to\infty} \sum_{k\in\mathbb{Z}} |\hat{f}_n(k)|^2 \\
&\le \lim_{n\to\infty} \int_{-\pi}^{\pi} |f_n(x)|^2 dx \\
&= \lim_{n\to\infty} \|f_n\|_{L^2}^2 \\
&= \|f\|_{L^2}^2
\end{aligned}$$

This shows that the Fourier series of any $f \in L^2$ is in ℓ^2 and $\|\mathcal{F}f\|_{\ell^2} \le \|f\|_{L^2}$. $\square$

Theorem 3.11. $\mathcal{F}$ is an isometry between L^2 and ℓ^2.

Proof. If $f \in L^2$, we take $f_n \in L^2$ that are continuous and have compact support in $(-\pi, \pi)$ that converge to f in L^2. We then have $|\,\|\mathcal{F}f\|_{\ell^2} - \|\mathcal{F}f_n\|_{\ell^2}\,| \le \|\mathcal{F}(f - f_n)\|_{\ell^2} \le C\|f - f_n\|_{L^2} \to 0$. By applying Parseval's identity, we get

$$\|\mathcal{F}f\|_{\ell^2} = \lim_{n\to\infty} \|\mathcal{F}f_n\|_{\ell^2} = \lim_{n\to\infty} \|f_n\|_{L^2} = \|f\|_{L^2}$$

This shows that $\mathcal{F}$ is an isometry, so it is automatically injective. For surjectivity, let $a_n \in \ell^2$ and define $f_n = \sum_{|k|\le n} e^{ikx} a_k$. For any $m > n$, we have

$$\|f_n - f_m\|_{L^2} = \|\mathcal{F}(f_n - f_m)\|_{\ell^2} = \left(\sum_{n<|k|\le m} |a_k|^2\right)^{1/2}$$

Since $\sum_{k=-\infty}^{\infty} |a_k| < \infty$, the above shows that f_n is a Cauchy sequence. Since L^2 is a Banach space, $f_n \to f \in L^2$, and then $\hat{f}(k) = \lim_{n\to\infty} \hat{f}_n(k) = a_k$. $\square$

Corollary 3.12. If $f \in L^2$ then $f_n = \sum_{|k|\le n} e^{ikx}\hat{f}(k)$ converges to f in L^2.

Proof.

$$\|f - f_n\|_{L^2}^2 = \|\mathcal{F}(f - f_n)\|_{\ell^2}^2 = \sum_{|k|>n} |\hat{f}(x)|^2 \to 0$$

$\square$